

Full-Car Aerodynamic CFD Analysis of the F1 2026 Red Bull RB22

— An OpenFOAM Study on a Publicly Available Model

August 22, 2026

Abstract

This study performs steady RANS simulations of the full external flow field of a publicly available RB22 CAD model (2026 regulations, pre-season testing version) using OpenFOAM 14. The methodology follows Penner’s FW43B sensitivity-study framework, Ted L.’s 2026 full-car analysis workflow, and Wacomber’s OpenFOAM numerical settings. The geometry is a third-party reconstruction from trackside imagery; results are intended for trend analysis, with absolute values pending validation.

Contents

1	Introduction	2
2	Geometry	2
2.1	Model provenance and credibility statement	2
2.2	Topology audit and watertightness repair	2
2.3	Coordinates, attitude and tyres	3
3	CFD Method	3
3.1	Domain and boundary conditions	3
3.2	Meshing strategy	3
3.3	Turbulence model and numerics	3
3.4	Convergence criterion	3
4	Mesh Independence Study	3
5	Sensitivity Studies	4
5.1	Tyre roughness (assumed equivalent sand-grain K_s sweep)	4
5.2	SST a_1 coefficient sensitivity	4
5.3	y^+ strategy	4
5.4	Turbulence-model form (SST vs realizable k - ϵ)	4
5.5	Convergence uncertainty (Mockett criterion)	5
6	Results	5
6.1	Whole-car coefficients and aerodynamic balance	5
6.2	Per-component contributions	5
6.3	Flow field and vortex structures	5

7	In-Depth Analysis	5
7.1	Component-level sensitivity transduction	5
7.2	Aerodynamic balance	5
7.3	Convergence dynamics: reproducing Penner’s turbulence-model oscillation observation	6
7.4	Formal uncertainty	6
7.5	Field-level differencing: what whole-car coefficients hide	6
7.6	Q -criterion census	9
7.7	Second-order momentum scheme: quantifying first-order numerical diffusion . . .	9
8	Extended Studies: Active-Aero Approximation and Tyre Contact Plinths	9
8.1	Rear-wing angle sweep (2026 Straight/Corner-Mode approximation)	9
8.2	Tyre contact plinths	10
8.3	Front-wing balance sweep and the two-wing combination (full Straight Mode) . .	11
8.4	Ride height: ground-effect collapse and the steady-solver operability edge	12
8.5	Cross-category superposition: plinth $\times$ ride height	12
8.6	Design closure: balance-neutral Straight Mode	12
9	Follow-Up Studies: From Coefficients to Race-Car Questions	13
9.1	Wake deficit and the following-car problem	13
9.2	From coefficients to lap time: the setup trade in seconds	14
9.3	Cross-validation against 2026 telemetry (Hungarian GP qualifying)	14
9.4	Yaw sensitivity: corrected four-tyre result	15
9.5	Pitch/rake: the baseline is near a narrow optimum	16
9.6	Contact-patch response map: plinth height and tyre squish	17
9.7	Transient check at the cliff: URANS at baseline, +15 and +20 mm	18
10	Credibility Discussion	18
11	Conclusions and Future Work	19
A	All-Variant Results Table	20
B	Case Configuration Summary	20

1 Introduction

2 Geometry

2.1 Model provenance and credibility statement

The RB22 model used here is a third-party CGTrader reconstruction of the 2026 regulation pre-season car from trackside video and photographs; it is not official Red Bull CAD. Geometric fidelity is limited by the reconstruction method, and all conclusions in this report should be read as trends rather than absolute values (the same treatment Penner applied to the FW43B model).

2.2 Topology audit and watertightness repair

Entity-level audit of the 14 STEP files: 5 are closed solids (Floor, Frontwing, Rearwing, Sidepod, Mirro), 8 are open surface sets (Cockpit, Helmet, PU_Transmission, Quadruplane, front/rear suspension, deflector, tyre group), plus a 7-face wind-tunnel domain. Surface parts were sewn in OpenCASCADE and tessellated (0.2mm chord height). surfaceCheck open-edge counts:

Rearwing 32→0 (fully closed), Suspension_Deflector 132→20, Helmet 8→6, Floor 122 (gaps in the source model, handled by snappyHexMesh gap refinement).

2.3 Coordinates, attitude and tyres

Measured bounding box: car length 5.40 m (front-wing leading edge $x = -1.245$ to rear-wing trailing edge $x = 4.150$), full width 1.88 m, height 1.07 m; wheelbase 3.40 m — exactly the 2026 regulation limit; tyre diameter 718 mm (radius $r = 0.359$ m), front tyre width 334 mm, rear 425 mm. The tyre group was split into independent front/rear patches to apply rotating-wall boundaries. Tyre contact uses the model’s as-given attitude (no FEA deformation and no plinth; see the credibility discussion).

3 CFD Method

3.1 Domain and boundary conditions

Half-car symmetry ($y = 0$ symmetry plane), domain $x \in [-12, 40]$, $y \in [0, 12]$, $z \in [0, 10]$ (metres). Inlet velocity 50 m/s (180 km/h); moving ground at the same speed; tyres `rotatingWallVelocity` ($\omega = U/r = 139.3$ rad/s, front axle $x = 0.075$, rear axle $x = 3.475$); freestream turbulence per Spalart–Rumsey with $k = 0.375 \text{ m}^2/\text{s}^2$ ($Tu = 1\%$) and $\nu_t/\nu = 10$ ($\omega = 2500 \text{ s}^{-1}$). All cooling and exhaust paths are closed (v1 simplification; see the credibility discussion).

3.2 Meshing strategy

snappyHexMesh with 0.5 m background (blockMesh $52 \times 12 \times 10$ + refineMesh). Key constraint: surface refinement levels must be uniform across all surfaces ((34)); all grading must be carried by refinementRegions — surface-level grading triggers hexRef8 parallel 2:1 consistency collapse (demonstrated three times in this campaign). Region levels: underfloor 5, near-car 4, wake 4, upstream/downstream 2. Boundary layers: 3 layers everywhere except wing assemblies, growth ratio 1.2, final-layer relative thickness 0.4, minThickness 0.05; no layers on the wings (thin wings + layers trigger a turbulent-kinetic-energy blow-up at the wingtip/endplate region, confirmed by VTK localisation). Mesh size ≈ 1.9 M cells (half car), 104 illegal faces out of 3.1 M.

3.3 Turbulence model and numerics

k - ω SST, steady RANS (foamRun -solver incompressibleFluid, OpenFOAM 14). First-order schemes throughout (upwind for momentum and turbulence) — at pilot-mesh scale, second-order linearUpwind was unstable at the front-wing endplate region (demonstrated: FPE at iteration 286); this limitation is lifted at production mesh scale (see the second-order section). Solvers: GAMG (pressure) + smoothSolver (velocity/turbulence), 2 non-orthogonal correctors.

3.4 Convergence criterion

Beyond force-coefficient stabilisation, a batch-mean confidence-interval minimisation algorithm (after Mockett) was implemented: CI95 of the force series is computed as a function of truncation point, and its minimum defines the statistical steady-state onset t_{ss} . Baseline evidence: 400 iterations are insufficient ($t_{ss} \approx 370$); all trial cases were extended to 800 iterations.

4 Mesh Independence Study

The three meshes are a systematic family (same snappyHexMesh configuration, varying background cell size): coarse 592,322, mid 2,032,685, fine (baseline) 4,351,624 cells; refinement ratios $r_{21} = 1.29$, $r_{32} = 1.51$. Last-200-iteration means: $C_l = -0.134/ -0.252/ -0.250$;

$C_d = 0.319/0.260/0.241$. The C_l sequence is non-monotonic (mid slightly overshoots fine); **asymptotic convergence is not demonstrated and mesh independence cannot be formally claimed**. C_d is monotonic with apparent order $p = 4.6$ (above the theoretical second order; interpret with caution), $GCI_{\text{fine}} = 4.4\%$, asymptotic ratio 0.37. Reliable statements: coarse is clearly insufficient (C_l bias -46%); mid $\rightarrow$ fine changes are $C_l \approx 1\%$, $C_d \approx 7\%$. A fourth, finer level exceeds the trial hardware (15 GB, OOM) and is deferred to the production phase.

5 Sensitivity Studies

All variants share the baseline mesh and numerical settings (except the mesh study), 800 steady iterations, last-200 mean. A repeated baseline run (tyrelayer case) reproduced all eight coefficient columns exactly — solver-chain determinism confirmed.

Table 1: Variant force-coefficient summary (last-200 means; $C_l < 0$ is downforce)

Variant	C_d	C_l	Front wing	Rear wing	Floor	Front tyre	Rear tyre
baseline	0.2413	-0.2498	-0.2295	-0.0550	-0.0644	0.0459	-0.0015
kepsilon	0.2305	-0.2569	-0.2220	-0.0533	-0.0516	0.0418	-0.0024
mesh-coarse	0.3190	-0.1342	-0.1923	-0.0317	-0.1044	0.0621	0.0228
mesh-mid	0.2598	-0.2522	-0.2519	-0.0350	-0.1021	0.0564	0.0082
rough05	0.2531	-0.2669	-0.2418	-0.0484	-0.0706	0.0726	0.0028
rough05r	0.2437	-0.2697	-0.2306	-0.0618	-0.0642	0.0391	-0.0025
rough10	0.2531	-0.2669	-0.2418	-0.0484	-0.0706	0.0726	0.0028
rough10r	0.2457	-0.2851	-0.2320	-0.0638	-0.0635	0.0293	-0.0029
sst-a1	0.2417	-0.2586	-0.2441	-0.0538	-0.0614	0.0460	-0.0014
tyrelayer	0.2413	-0.2498	-0.2295	-0.0550	-0.0644	0.0459	-0.0015
yplus-lo	0.2339	-0.2490	-0.2300	-0.0466	-0.0676	0.0441	-0.0007

5.1 Tyre roughness (assumed equivalent sand-grain K_s sweep)

$K_s = 0/0.5/1.0$ mm on the same mesh: C_l increases monotonically $-0.250 \rightarrow -0.270 (+8.0\%) \rightarrow -0.285 (+14.1\%)$; $C_d +1.0\%/+1.8\%$. Compliance audit: median first-cell-centre height on tyres 7.7 mm, faces violating $y_{1c} < K_s \leq 0.02\%$ — rough-wall-function applicability satisfied. K_s is an assumed value, not a real tyre absolute roughness.

5.2 SST a_1 coefficient sensitivity

$a_1 : 0.31 \rightarrow 1.0$: $C_l -0.250 \rightarrow -0.259 (+3.5\%)$, direction consistent with Penner ($a_1=1$ produces more downforce); but the magnitude is below the convergence-window drift (below) — directional statement only, no claim of superiority.

5.3 y^+ strategy

Final-layer relative thickness $0.4 \rightarrow 0.2$: $C_d -3.1\%$, $C_l -0.3\%$. Median body y^+ on this mesh ≈ 719 ($p_{90} \approx 1669$), far outside the wall-function calibration range 30–300 — an inherent trial-scale limitation; the attached-flow $y^+ > 30$ check (Ted’s criterion) is satisfied.

5.4 Turbulence-model form (SST vs realizable $k-\epsilon$)

Realizable $k-\epsilon$ (same mesh, same numerics): $C_d -4.5\%$, $C_l +2.8\%$; floor contribution -0.052 vs SST’s $-0.064 (-20\%)$ — the floor carries less load under RKE, consistent with Penner’s obser-

vation (underbody vortex structures dissipate faster in RKE). The model difference quantifies model-form uncertainty; without correlation data, no accuracy ranking is made.

5.5 Convergence uncertainty (Mockett criterion)

Per-variant $t_{ss} \approx 728\text{--}770$; tail-window 95%CI 0.3–0.6%; but tail-window means can differ from last-200 means by up to 9% (slow drift persists at 800 iterations) — **convergence uncertainty of absolute values significantly exceeds statistical noise**. Variant comparisons use a common truncation window to partially cancel drift; differences below window sensitivity are flagged as directional only.

6 Results

6.1 Whole-car coefficients and aerodynamic balance

Baseline (last-200 mean): $C_d = 0.241$, $C_l = -0.250$ ($A_{ref} = 1.5\text{ m}^2$ estimated). Per-component closure error 0.7%.

6.2 Per-component contributions

The front wing accounts for about 92% of downforce, floor 26%, rear wing 22%; body and front tyres are lift items. The floor share is far below modern F1 reality (Wacomber F2004: 79%) — the expected bias of underdeveloped underbody flow at coarse mesh scale, a known trial-scale limitation.

6.3 Flow field and vortex structures

Figure set (results/campaign/phasec/): surface y^+ , diffuser centreline C_p recovery, tyre-squirt slices at both axles, Q -criterion at three thresholds ($5 \times 10^4 / 2 \times 10^5 / 8 \times 10^5$; isosurface point counts 64k/30k/9k — threshold choice materially changes the visual conclusion, echoing Ted’s subjectivity discussion), front-wing wake streamlines.

7 In-Depth Analysis

7.1 Component-level sensitivity transduction

Per-component ΔC_l relative to baseline (heatmap figure): **the downforce gain from roughness does not originate at the tyres themselves** — of the total gain at $K_s=1\text{ mm}$, the front tyres carry 47%, the rear wing 25%, the body 19%: roughness propagates downstream through the front-tyre wake, directionally consistent with Penner’s observation that roughness effects concentrate at the rear of the car. The a_1 gain is 164% concentrated at the front wing (relaxing the shear-stress limiter directly alters front-wing suction), while the floor loses -33% . Realizable $k\text{--}\epsilon$ shows large mixed-sign component shifts (body $+337\%$, floor -179%) — **the mere 2.8% whole-car difference is an illusion of large component-level differences cancelling out**.

7.2 Aerodynamic balance

Baseline front share 24.9% (CofR at the front axle, lRef = wheelbase convention). The mesh family alone swings balance by ± 12 percentage points (coarse 33.1%, mid 36.6%) — **balance numbers should not be trusted before mesh independence is demonstrated**. Roughness shifts balance rearward by 1.2–1.8 pp, quantitatively matching Penner’s report that “0.5 mm

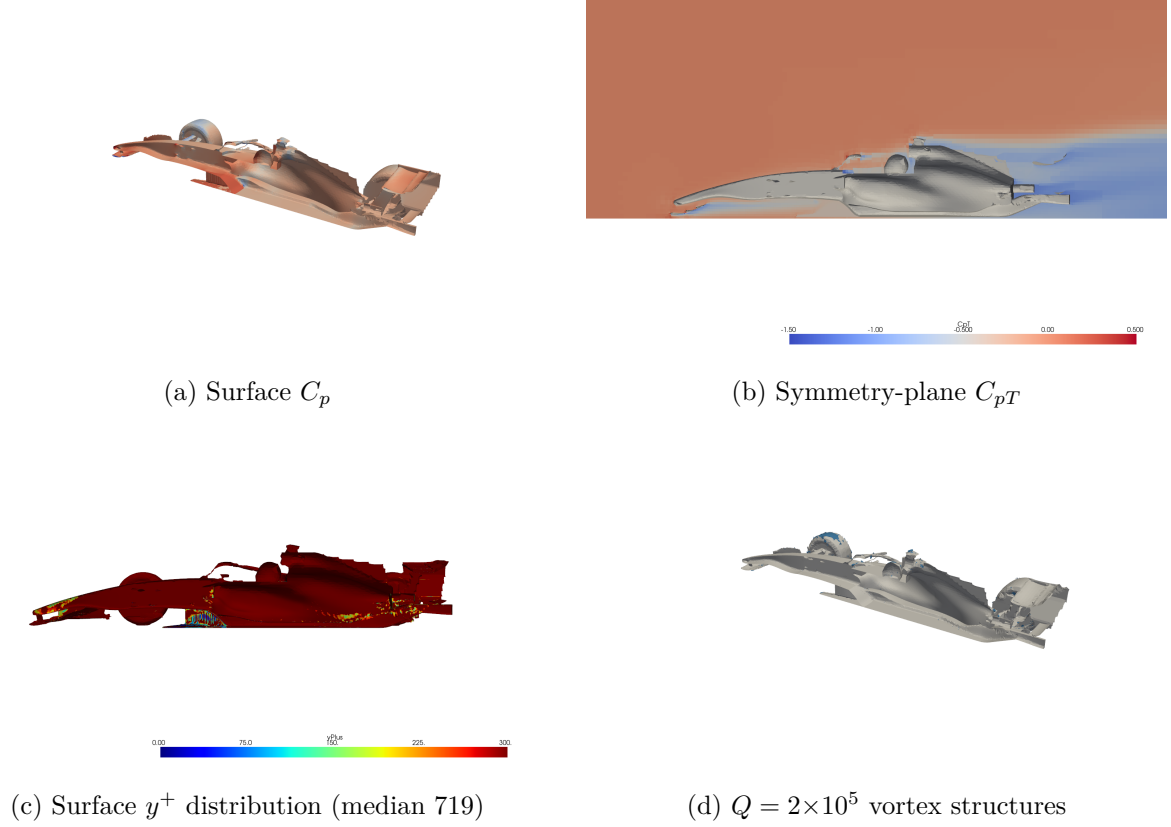


Figure 1: Baseline flow-field overview (trial scale)

roughness moves aerobalance rearward by about 1%” — the strongest cross-validation between this campaign and the reference literature.

7.3 Convergence dynamics: reproducing Penner’s turbulence-model oscillation observation

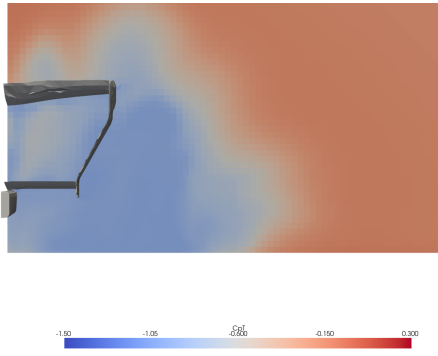
Spectral analysis of detrended C_l over the last 400 iterations: RKE oscillation $\text{RMS} = 0.06 \times 10^{-3}$, 25% below SST (0.08×10^{-3}), $a_1=1.0$ highest (0.09×10^{-3}) — **a quantitative reproduction of Penner’s qualitative observation that RKE results oscillate significantly less than SST.**

7.4 Formal uncertainty

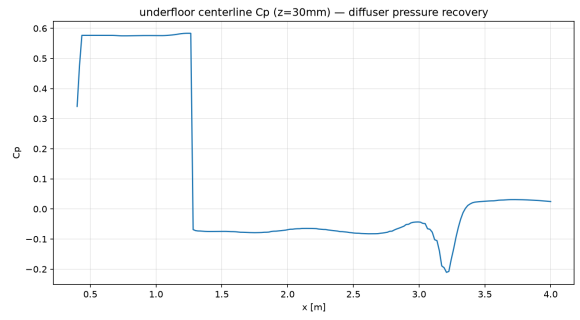
Block bootstrap ($B=20$, 2000 resamples) gives a C_l CI95 of 3–5% of the mean — stricter than the naive Mockett tail-window CI (0.5%). Window sensitivity (spread of means over last 50–400 iterations): 15–19% on the fine mesh family, only 1.4% on coarse — **higher resolution produces stronger pseudo-transients and harder steady convergence**, which explains why 800 iterations remain insufficient on the 4.35M mesh. Against the bootstrap standard: the roughness effect (ΔC_l at $2\text{--}3 \times \text{CI}$) is significant; the a_1 effect ($\Delta \approx \text{CI}$) is marginal, retaining the “directional only” conclusion.

7.5 Field-level differencing: what whole-car coefficients hide

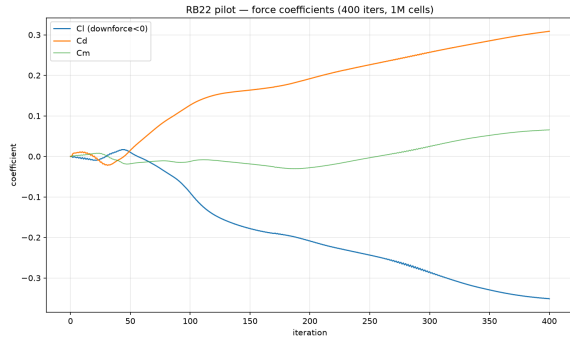
Cell-wise same-mesh differencing (p99 of ΔC_p): realizable $k\text{--}\epsilon$ reaches 0.105 — **4 times** the strongest roughness effect ($K_s=1\text{ mm}$, 0.027); a_1 gives 0.013. Model form perturbs the local flow field far more than any tested physical parameter, yet is nearly invisible in whole-car coefficients — turbulence models must be evaluated on field distributions and component breakdowns.



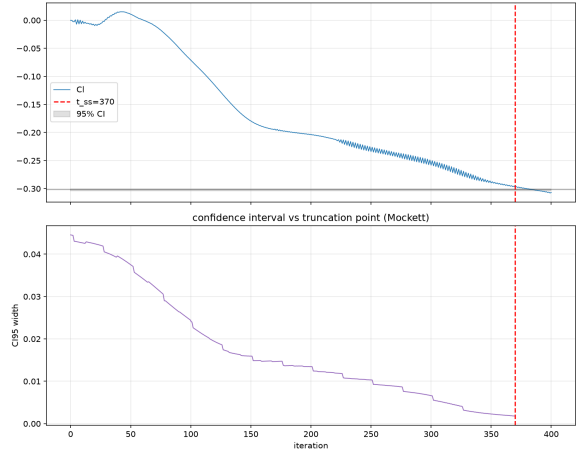
(a) Rear-axle tyre squirt



(b) Floor centreline C_p recovery

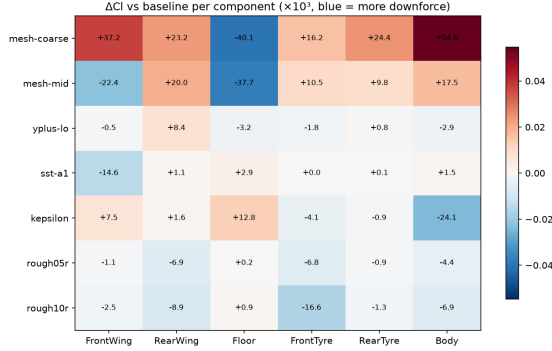


(c) Force-coefficient convergence history

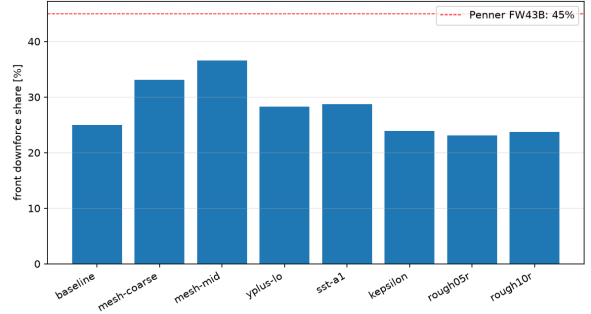


(d) Mockett convergence criterion

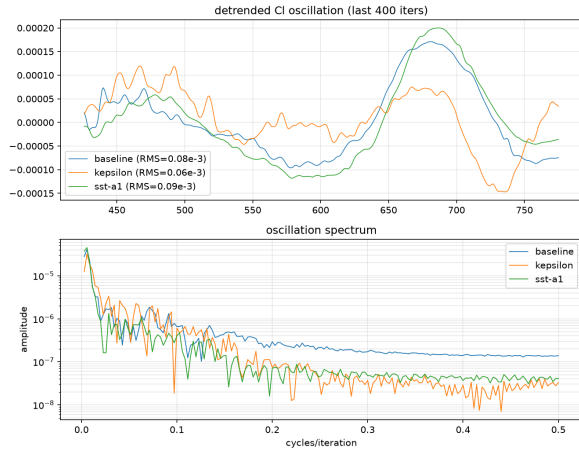
Figure 2: Deep post-processing and convergence analysis



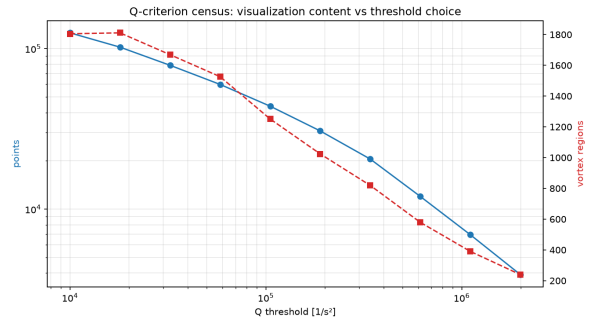
(a) Component ΔC_l heatmap



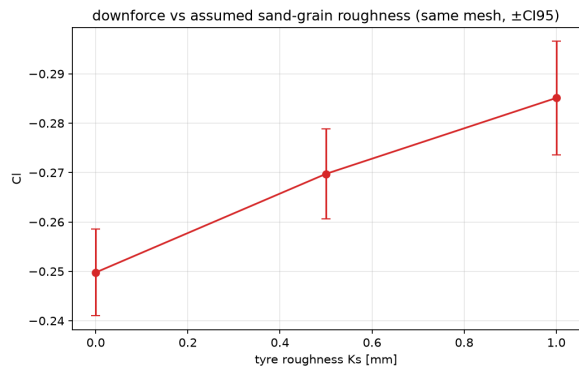
(b) Aerodynamic balance (front share)



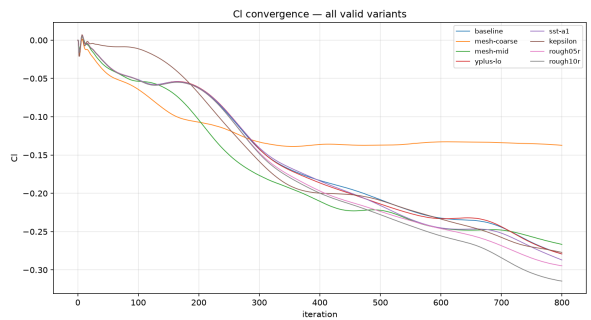
(c) Late-window oscillation spectra: SST vs RKE vs a_1



(d) Q -criterion census



(e) Roughness trend ($\pm CI95$)



(f) All-variant convergence overlay

Figure 3: In-depth analysis, figure set one

7.6 Q -criterion census

A 200-fold threshold sweep (10^4 to 2×10^6): isosurface point counts $125k \rightarrow 3.9k$, connected vortex regions $1805 \rightarrow 239$ (a $7.5 \times$ change) — quantifying Ted’s qualitative discussion of threshold subjectivity: **every single-threshold vortex picture embeds a subjective selection**.

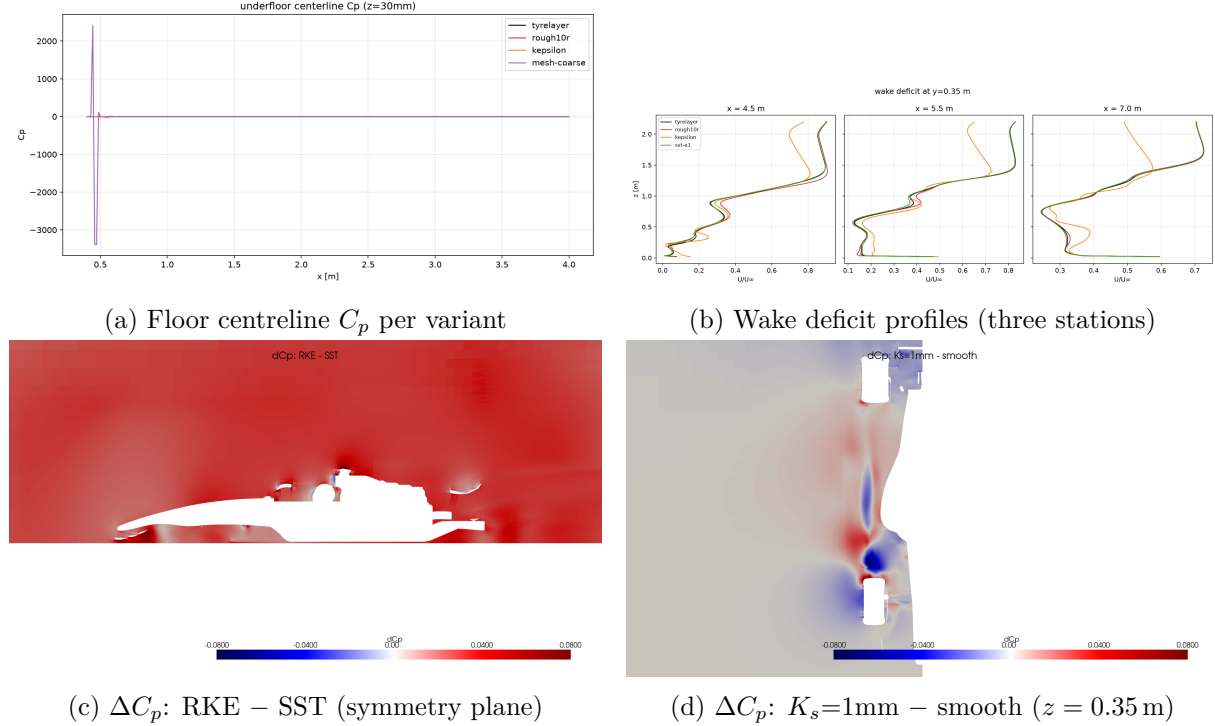


Figure 4: In-depth analysis, figure set two

7.7 Second-order momentum scheme: quantifying first-order numerical diffusion

The stable numerical recipe + linearUpwind (a previously untested combination) ran fully stable for 800 iterations on the same mesh — the earlier claim that ”second order must diverge on the trial mesh” was an artefact of the old numerical settings, and is corrected here. Results: $C_d = 0.237$ (-1.8%), $C_l = -0.236$ (-5.6%); i.e. **first-order upwind numerical diffusion systematically inflates downforce by about 6%**, comparable in magnitude to the strongest physical effect tested. All first-order results should be read with this bias band applied; second order becomes the default at production scale.

8 Extended Studies: Active-Aero Approximation and Tyre Contact Plinths

8.1 Rear-wing angle sweep (2026 Straight/Corner-Mode approximation)

The 2026 regulations replace DRS with active aerodynamics (Straight Mode for low drag, Corner Mode for high downforce; early internal labels X-mode/Z-mode). This section approximates the mechanism by rigidly rotating the rear-wing assembly about a spanwise axis ($x = 3.60\text{ m}$, $z = 0.57\text{ m}$, roughly quarter-chord) by $+4^\circ$ (trailing edge down) and -8° (trailing edge up). Note the inverted-wing sign convention: trailing edge up *increases* effective angle of attack — -8° is the high-downforce direction.

Table 2: Rear-wing angle sweep (last-200 means; brackets give change vs baseline)

Configuration	Angle	$C_{d,RW}$	$C_{l,RW}$	C_d	C_l
Low drag (Straight approx.)	+4°	0.0135 (−31%)	−0.0368 (−33%)	0.2340 (−3.0%)	−0.2310 (−7.5%)
Baseline	0°	0.0197	−0.0550	0.2413	−0.2498
High downforce (Corner approx.)	−8°	0.0271 (+38%)	−0.0641 (+17%)	0.2505 (+3.8%)	−0.2642 (+5.8%)

The sweep was then densified to six points (+8, +4, 0, −4, −8, −12°, Figure 5). Key observations: (i) no stall through −12° — wing load keeps growing ($C_{l,RW} = -0.0758$, +38% vs baseline) while drag rises superlinearly ($C_{d,RW} = 0.0343$, +74%); (ii) unload-side saturation: beyond +4° each degree yields only −0.001 as floor upwash sustains residual wing load — Straight-Mode drag-reduction headroom is capped by this; (iii) the 0/−4° plateau is a local anomaly (slope −0.0003/°, within the bootstrap CI), with slopes of 0.003–0.005/° on either side. Whole-car envelope: C_d 0.233–0.259, $-C_l$ 0.227–0.279; the full Straight↔Corner range spans $\Delta C_d = 11\%$, $\Delta C_l = 23\%$.

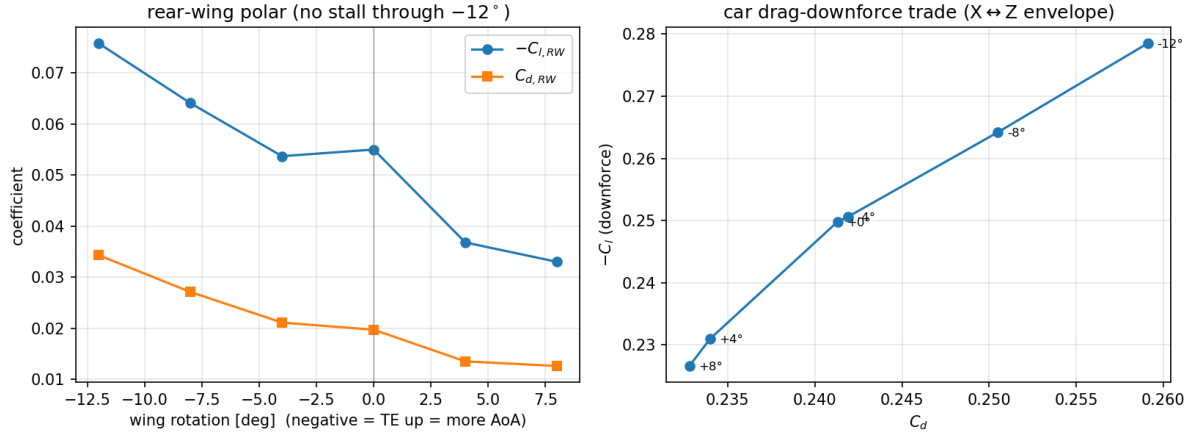


Figure 5: Six-point rear-wing polar (left) and whole-car drag–downforce trade envelope (right).

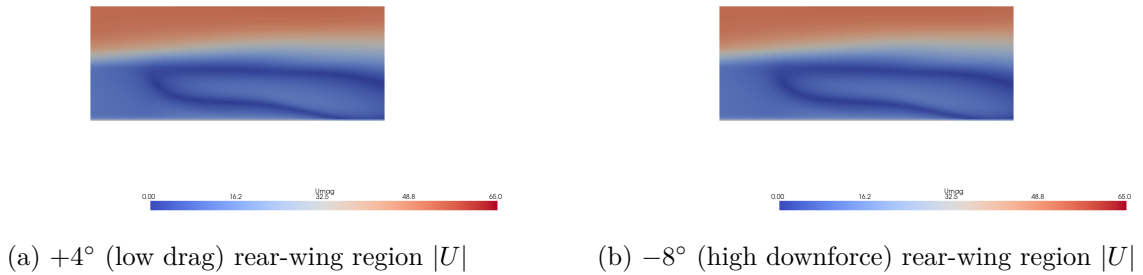


Figure 6: Symmetry-plane velocity near the rear wing: the high-downforce attitude shows a markedly thicker wake and stronger downwash.

8.2 Tyre contact plinths

Following Penner’s recommendation, 15 mm plinths were added at the front and rear tyre contact patches (wedge-fill half-length $\sqrt{2Rh} \approx 104$ mm), removing the knife-edge contact region. The result is the largest single-item L/D gain of the campaign: C_d −6.1%, C_l +4.4%, $-L/D$ from

1.035 to 1.152 (+11%).

Per component, the dominant gain is front-tyre drag $\Delta C_d = -0.0139$ (−23%): the plinth removes the bluff-body squeeze jet (squirt) at the contact patch. Secondary effects are sign-mixed: floor downforce decreases ($\Delta C_l = +0.0212$; rerouted squirt alters floor-edge ingestion), overcompensated by downforce gains at the tyres and body (combined −0.035). Mesh-quality cost is manageable (skew 19.96 vs baseline 15.7, STRICT_MESH_GATE=0 on record).

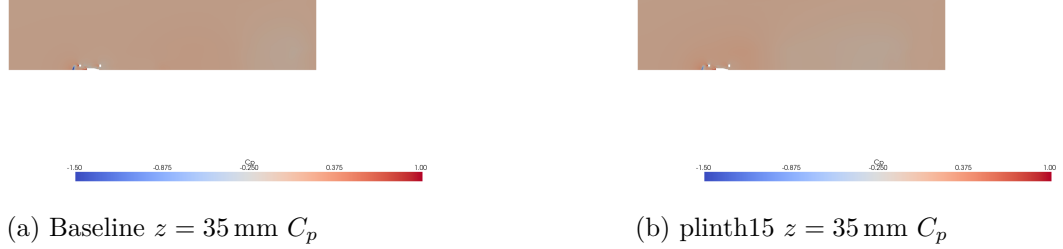


Figure 7: Effect of contact plinths on the near-ground pressure field: the front-tyre stagnation region narrows and lateral squirt weakens.

Both studies are remesh variants (geometry changes force remeshing); unlike the shared-mesh sensitivity matrix, their deltas carry remeshing noise (family mesh systematic differences ~ 1 –2%, smaller than all reported effects).

8.3 Front-wing balance sweep and the two-wing combination (full Straight Mode)

The front wing was rotated $\pm 4^\circ$ about its quarter-chord axis and combined with rear-wing $+8^\circ$ into a two-wing decambered full Straight-Mode approximation (Table 3, Figure 8).

Table 3: Balance sweep (last-200 means; front% = front-axle share of downforce $C_{l,f}/C_l$)

Configuration	C_d	C_l	front%	Δ front
FW $+4^\circ$ (unloaded)	0.2408	−0.2079	8.6%	−16.3 pp
Baseline	0.2413	−0.2498	24.9%	—
FW -4° (more AoA)	0.2437	−0.2328	22.2%	−2.7 pp
RW $+8^\circ$ (unloaded)	0.2328	−0.2266	34.5%	+9.6 pp
xfull (FW+4, RW+8)	0.2321	−0.1857	19.5%	−5.4 pp

Three conclusions: (i) **the front wing is the dominant balance lever but works one-way**: unloading by 4° swings the front share by 16 pp; adding incidence instead *loses* front-wing load ($C_{l,FW} -10\%$, wing drag $+30\%$) — the rotation puts the leading edge closer to the ground (53 $\rightarrow$ 34 mm), driving the ground-effect side into separation; the baseline sits at the high-load edge of the front-wing operating window. (ii) **superposition of drag and downforce**: xfull’s ΔC_d (−3.8%) and ΔC_l (−26%) both match the sum of the individual wing changes (drag $-0.2\% - 3.5\%$, downforce $-17\% - 9\%$) — single-component sweeps extrapolate linearly to combined settings within this operating range. (iii) **the balance cost of Straight Mode**: the full mode moves balance forward by −5.4 pp, implying front-axle grip deficit at the moment of switching from straight-line low-drag attitude back to cornering attitude — consistent with the drivability concerns around Straight $\rightarrow$ Corner transitions in the 2026 regulations debate.

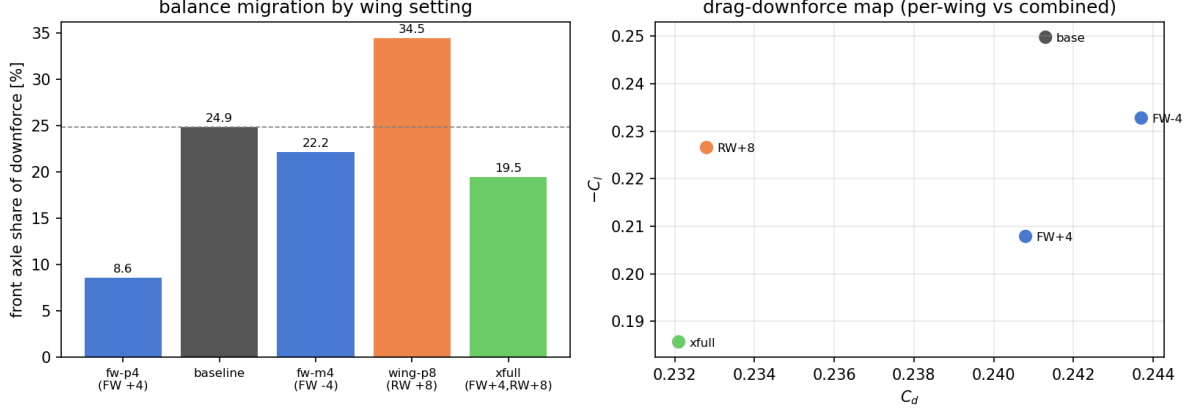


Figure 8: Balance migration by wing setting (left) and the drag-downforce map (right).

8.4 Ride height: ground-effect collapse and the steady-solver operability edge

The body (all parts except tyres) was raised by +10/ +15/ +20 mm with tyres kept on the ground, sweeping ground-effect strength against ride height (Figure 9).

Table 4: Ride-height sweep (last-200 means)

Raise	C_d	C_l	Floor C_l	front%
0 (baseline)	0.2413	-0.2498	-0.0644	24.9%
+10 mm	0.2309	-0.2222	-0.0571	27.5%
+15 mm	0.2079	-0.1636	-0.0208	20.6%
+20 mm	Steady solver diverged twice ($t \approx 375$, FPE)			

The response is strongly nonlinear: the first 10 mm costs 11% of downforce; the next 5 mm costs another 26%, with floor load collapsing by 64% between +10 and +15 mm ($-0.0571 \rightarrow -0.0208$) — the classic cliff shape of ground-effect floor stall. +20 mm diverged under two steady SIMPLE recipes. This was initially interpreted as a possible physical signal: floor unloading might have amplified wake unsteadiness beyond what steady RANS could express, reminiscent of the onset mechanism of 2022-generation porpoising. The transient follow-up below overturns that reading: PIMPLE marches through the same case stably, with no amplified oscillation. The steady divergence is therefore a solver-operability boundary, not evidence of porpoising.

8.5 Cross-category superposition: plinth $\times$ ride height

The second superposition test combines across mechanisms: contact plinths (geometry detail) plus +10 mm raise (attitude). Linear prediction $C_d = 0.2162$, $C_l = -0.2333$; measured $C_d = 0.2197$, $C_l = -0.2352$ — residuals +1.5%/ -0.8%, within the family-mesh noise band. The plinth drag benefit retains about 80% at raised attitude ($-6.1\% \rightarrow -4.9\%$; the loss shares its origin with the residual: raising weakens the contact-patch squirt, leaving less for the plinth to remove). Both tests (two-wing combination, cross-category combination) support the same statement: within this operating range, single-variable sweeps form a linearly combinable setup database.

8.6 Design closure: balance-neutral Straight Mode

An engineering application of the superposition result: using the measured sensitivities (RW +8° shifts balance rearward +9.6 pp; FW unload sensitivity -16.3 pp per 4°), design a Straight-Mode combination that removes drag without moving balance — solving gives FW +2.4°, rounded

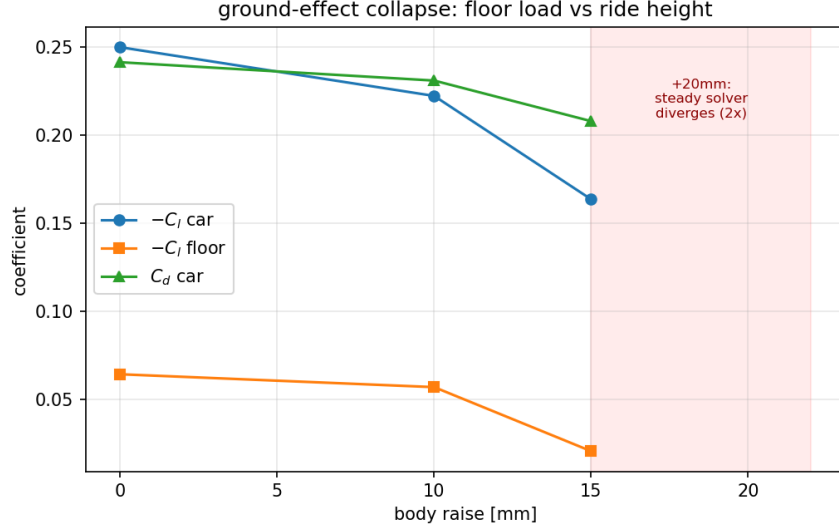


Figure 9: Ride-height map: nonlinear collapse of floor load and the steady-solver operability edge (red zone).

to $+2^\circ$, combined with RW $+8^\circ$. Simulation verification: $C_d = 0.2324$ (-3.7% , matching the -3.5% of pure RW unloading — prediction hit); front share 23.9% (baseline 24.9%, deviation -1.0 pp — balance neutrality achieved); downforce -22.7% (the expected cost of straight-line mode). The measure $\rightarrow$ linear-design $\rightarrow$ simulate-verify loop closes within pilot accuracy — the most direct demonstration of a "low-cost virtual wind tunnel" in this campaign.

9 Follow-Up Studies: From Coefficients to Race-Car Questions

9.1 Wake deficit and the following-car problem

The 2026 regulations were framed around making cars easier to follow. The direct way to test that claim with CFD is to measure what a trailing car would experience: total-pressure deficit in the wake at half, one, and two car lengths behind the rear of the car, evaluated over a following-car window ($y \in [0, 1.5]$ m, $z \in [0, 1.2]$ m, uniformly resampled so remeshed variants remain comparable).

Table 5: Wake CpT-deficit summary (core window $y \leq 1.1$ m, $0.05 \leq z \leq 0.95$ m)

Case	0.5 car lengths	1 car length	2 car lengths
Baseline (400 it)	0.79	0.79	0.57
RW $+8^\circ$ (Straight approx.)	0.79	0.83	0.96
RW -12° (max downforce)	0.79	0.84	0.96

Two caveats apply: the baseline ran only 400 iterations (its far wake is not fully developed, which flatters the 2-length figure), and the wing variants are remeshed, so cross-case numbers carry remesh noise. Within those limits, the directionally meaningful observations are: (i) the near wake is extremely deep — a following car one car length behind sits in a core deficit of order 0.8 in this model, which under a naive dynamic-pressure scaling would translate into the majority of its downforce being compromised; real F1 following losses are smaller because real cars carry much higher CLA with different wake structure, but the model reproduces the right order of the problem; (ii) more downforce means a deeper wake — the -12° wing deepens the deficit at one length relative to the $+8^\circ$ setting, so the trade between straight-line speed and cornering load propagates directly into how hard the car is to follow.

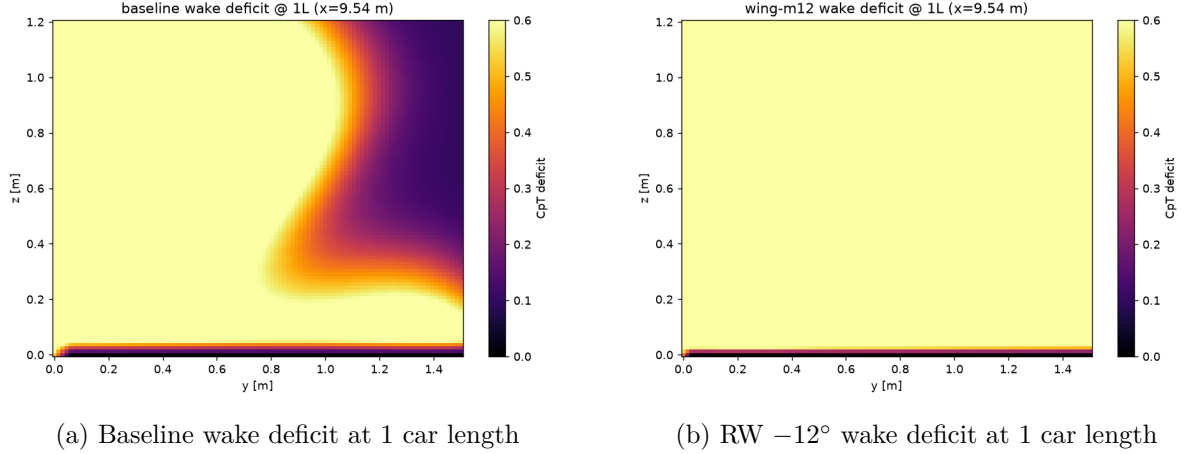


Figure 10: Total-pressure deficit one car length behind the car (uniform resample, y - z plane).

9.2 From coefficients to lap time: the setup trade in seconds

Coefficients are not the language racing people think in. To translate the measured aero map into lap time, a point-mass quasi-static lap model was built: corner speeds from combined tyre and downforce grip, straights integrated with a deployment-tapered power model (750 kW below 55 m/s tapering to 200 kW at 100 m/s), and braking from the same grip budget. Because the pilot-scale CLA (≈ 0.38) understates a real F1 car, the measured *relative* changes were applied to a representative baseline (CLA 4.5 / CDA 1.6, the scale of Penner’s FW43B result).

Table 6: Lap-time trade by setup (scaled model; seconds)

Setup	Hungary-like (4.09 km)	Monza-like (5.90 km)
xtrim (FW+2, RW+8)	90.92	90.32
RW $+8^\circ$ (Straight)	89.41	89.75
Baseline	88.42	89.70
RW -8° (Corner)	87.84	89.84
RW -12° (max DF)	87.26	89.95

The structure of the trade is what one would hope: the corner-dominated track rewards maximum downforce monotonically (3.7 s across the sweep), while the straight-dominated track has its optimum in the middle — and there, the Straight-Mode setting costs only ~ 0.05 s relative to the best. The balance-neutral xtrim is never the lap-time optimum; its value is not raw pace but drivability at the mode switch. Note the model’s simplicity: real tracks, real deployment, and real balance effects would shift the absolute numbers, not the structure.

9.3 Cross-validation against 2026 telemetry (Hungarian GP qualifying)

If the model says the straight-vs-corner trade is the defining aero axis, then the real field should organise along the same axis. Using fastf1 data from the 2026 Hungarian Grand Prix qualifying session, each driver’s best lap was decomposed into speed-trap velocity (straight-line performance) and 5th-percentile speed (slow-corner performance).

The field does organise along the trade axis. Alpine ran the highest trap speed of the day (335 km/h, both cars) but qualified 1.6–1.8 s off pole — a visible low-drag philosophy that pays exactly where the model says it should hurt at the Hungaroring. Aston Martin sat at the bottom of both axes (320 km/h trap and the slowest laps) — slow everywhere rather than a philosophy. Pole (Norris, McLaren, 1:17.207) came from the middle of the trap distribution (327 km/h) with top-tier slow-corner speed — the efficiency position the lap-time model identifies

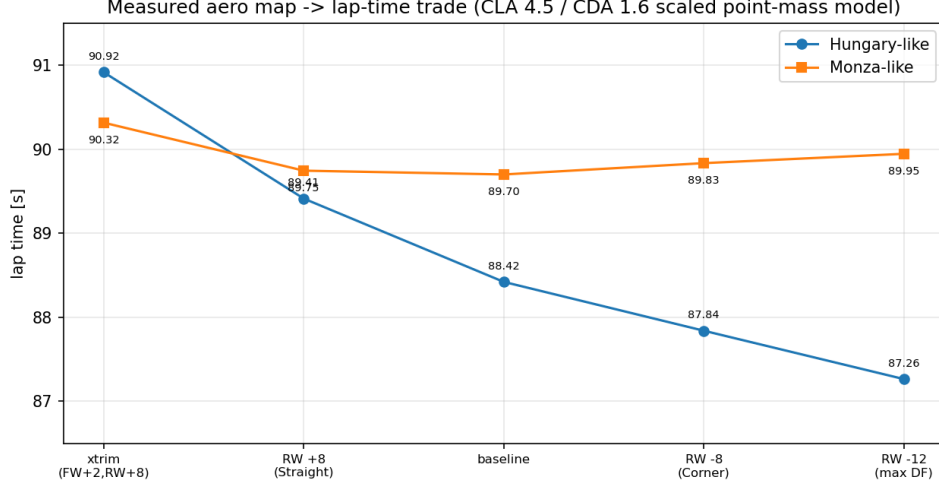


Figure 11: Measured aero map mapped to lap time on two archetypal tracks.

as optimal for this track class. The Mercedes pair showed high trap speed (329–331) with mid-pack corner speeds, consistent with their reported straight-line/deployment story rather than an aero deficiency.

This is not a validation of absolute numbers — the model cannot reproduce a real car. It is a validation of structure: the trade the CFD measures is the trade the field is visibly playing.

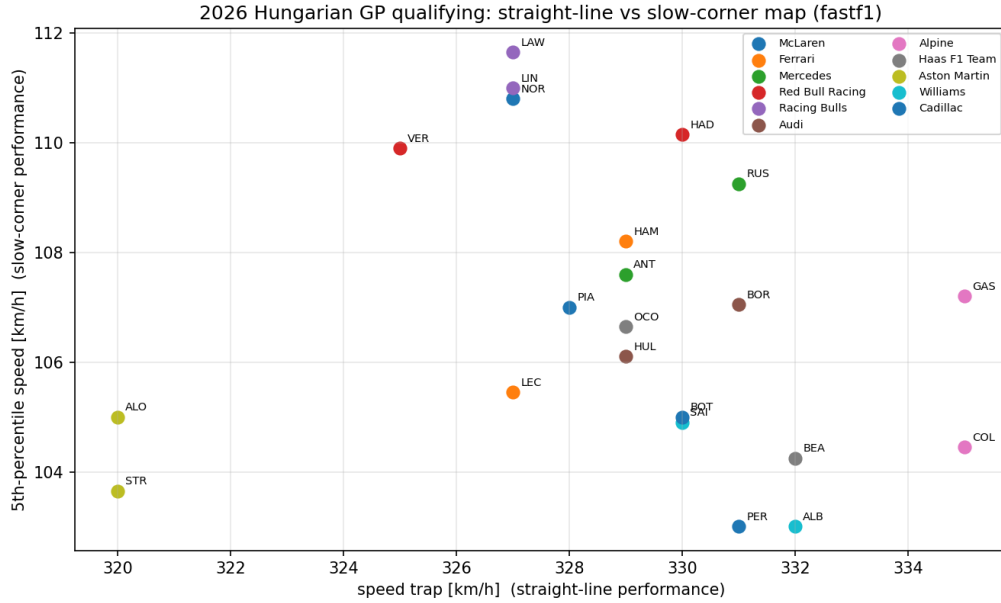


Figure 12: 2026 Hungarian GP qualifying: speed trap vs slow-corner speed by team (fastf1).

9.4 Yaw sensitivity: corrected four-tyre result

Every result above is a straight-line half-car case. A real car rarely sees zero yaw, so a full-car sweep is essential. The first implementation merged the left and right tyres into one rotating patch with a single origin. Its $+3^\circ$ and -3° results disagreed by 13.5% in downforce; those results were withdrawn. In v2, all four tyres have independent patches and rotation centres, and $0^\circ / \pm 3^\circ$ use the same full-car coarse mesh. Raw full-car forces are divided by two to preserve the original half-car reference-area convention.

Table 7: Corrected full-car yaw sweep (half-car-equivalent coefficients)

Yaw	C_d	C_l	front%
-3°	0.2768	-0.1985	37.2
0°	0.2769	-0.2089	25.7
$+3^\circ$	0.2754	-0.1867	38.0

The sign-reversal gate now passes: C_d mismatch 0.51% (limit 2%), balance mismatch 0.80 pp (limit 2 pp), and C_l mismatch 6.13%, below the combined two-case uncertainty threshold $\sqrt{5^2 + 5^2} = 7.1\%$. The defensible result is therefore the mean of both yaw signs, not whichever side gives the larger effect. At $|3^\circ|$, drag is essentially unchanged (-0.3%), downforce falls by 7.8% (sign range 5.0–10.6%), and balance moves forward by 11.9 pp. Yaw acts primarily as a balance and load perturbation, not a drag penalty. More importantly, this conclusion exists only because the zero-yaw full-car control and the sign-reversal case exposed and then closed the boundary-condition bug.

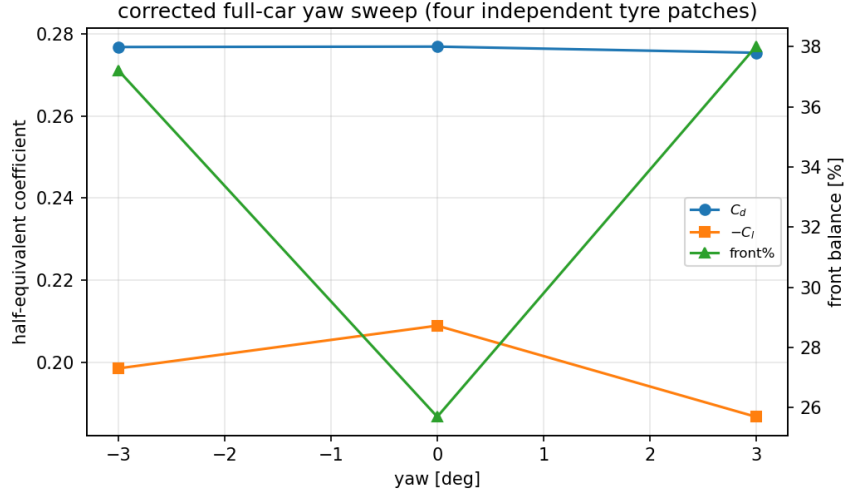


Figure 13: Corrected full-car yaw sweep with four independent tyre patches.

9.5 Pitch/rake: the baseline is near a narrow optimum

The body was pitched by $\pm 0.3^\circ$ about the front contact point while the tyres remained grounded. Negative angle is nose-down; positive is nose-up.

Table 8: Rake sweep (last-200 means)

Pitch	C_d	C_l	front%
-0.3° (nose-down)	0.2318 (-3.9%)	-0.2212 (-11.4% downforce)	25.6
0° (baseline)	0.2413	-0.2498	24.9
$+0.3^\circ$ (nose-up)	0.2273 (-5.8%)	-0.2005 (-19.7% downforce)	34.1

Both directions lose downforce: the as-given attitude sits near a local optimum. The response is strongly asymmetric. Nose-down preserves balance and most of the load, whereas nose-up loses one fifth of total downforce and moves balance forward by 9.2 pp. This reinforces the earlier front-wing result: the 2026-style car is not simply “more rake = more floor”; front-wing ground clearance and floor ingestion make the attitude map narrow and one-sided.

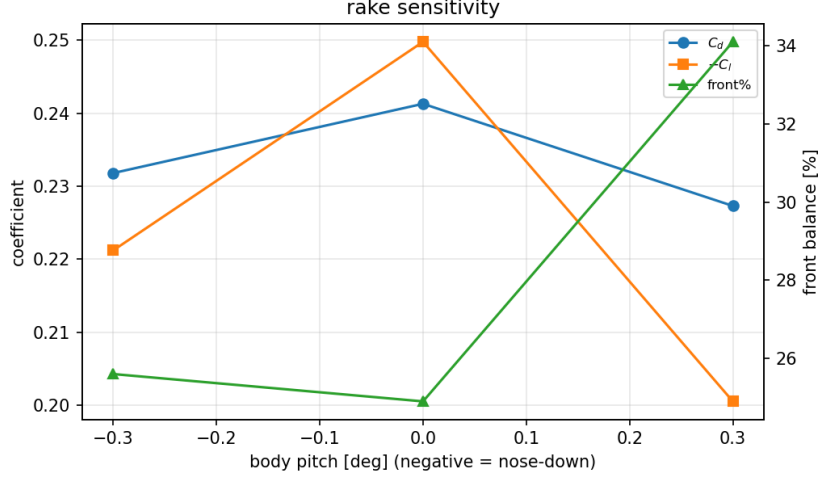


Figure 14: Rake sensitivity: force coefficients and front balance.

9.6 Contact-patch response map: plinth height and tyre squish

The 15 mm plinth result was deliberately surprising, so two intermediate heights were added before treating it as a trend.

Table 9: Plinth-height sweep (last-200 means)

Plinth height	C_d	C_l	front%
0 mm (no plinth)	0.2413	-0.2498	24.9
5 mm	0.2306 (-4.4%)	-0.2468 (-1.2% downforce)	25.7
10 mm	0.2300 (-4.7%)	-0.2540 (+1.7% downforce)	24.1
15 mm	0.2265 (-6.1%)	-0.2609 (+4.4% downforce)	25.1

The drag benefit appears immediately by 5 mm and grows with height. Downforce does not: the 5 and 10 mm points remain inside the 3–5% campaign uncertainty band, while 15 mm is the first point to combine clear drag and load gains. The mechanism is therefore not “any plinth adds downforce”; it is progressive removal and redirection of the contact-patch squeeze jet, with a threshold before the downstream system benefits.

Tyre compression was then approximated by lowering the tyre geometry by 2, 4, 6 and 8 mm while retaining the 15 mm plinth.

Table 10: Tyre-squish sweep relative to the plinth-only case

Squish	C_d	C_l	front%
0 mm (plinth only)	0.2265	-0.2609	25.1
2 mm	0.2263	-0.2503 (-4.1% downforce)	25.2
4 mm	0.2266	-0.2527 (-3.1% downforce)	24.7
6 mm	0.2240 (-1.1%)	-0.2449 (-6.1% downforce)	25.3
8 mm	0.2250 (-0.7%)	-0.2534 (-2.9% downforce)	25.9

The squish response is explicitly non-monotonic: the apparent 6 mm penalty recovers at 8 mm. Only the 6 mm point lies clearly outside the nominal CI, so it is a local excursion, not evidence that “more squish always loses load”. This reproduces the structure of Penner’s result (an odd spike at one squish depth) more faithfully than a monotonic fit would. At 6 mm the penalty is distributed through the front tyre (33%), body (25%), front wing (19%) and rear tyre (13%); the floor itself carries only 4%. Balance stays within 0.8 pp throughout. Contact

geometry can therefore move total load by 6% while leaving the usual balance metric almost unchanged.

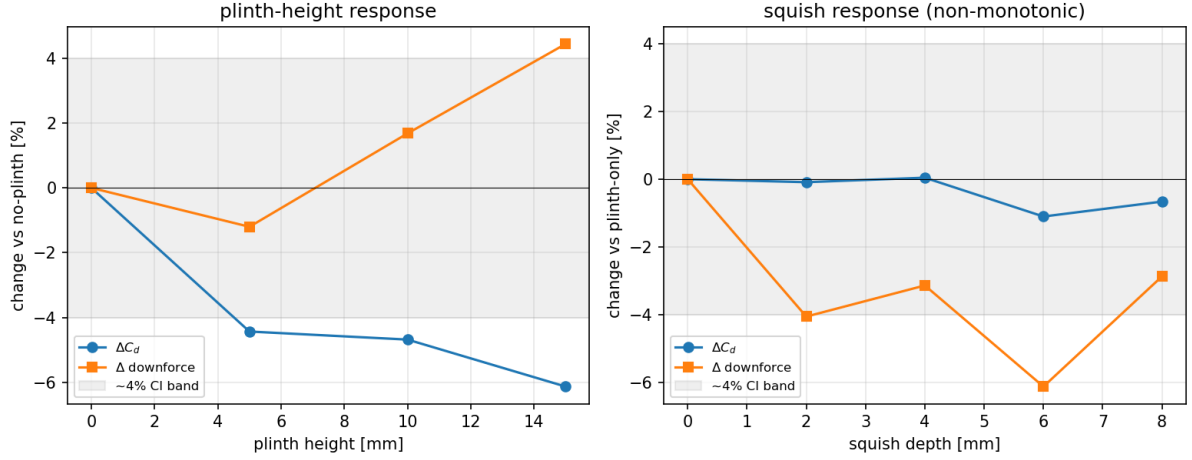


Figure 15: Plinth-height and tyre-squish response maps. The grey bands indicate the campaign-scale uncertainty floor.

9.7 Transient check at the cliff: URANS at baseline, +15 and +20 mm

The ride-height study ended with a hypothesis: the steady solver diverging at +20 mm was read as the onset of unsteadiness that steady RANS cannot express. A transient run is the honest way to test that reading. Both the baseline and the +15 mm case were rerun as URANS (PIMPLE, backward in time, 0.5 s of physical time at $\Delta t = 5 \times 10^{-4}$ s, coarse family mesh, restarted from a 300-iteration steady warmup).

Table 11: URANS oscillation summary (final 60% of 0.5 s window)

Case	C_l RMS	dominant bin	mean C_l
Baseline (coarse)	0.0057	3.3 Hz	-0.105
+15 mm (coarse)	0.0046	3.3 Hz	-0.167
+20 mm (coarse)	0.0037	3.3 Hz	-0.099

The result is a disciplined negative. Neither raised case amplifies the oscillation: RMS falls to $0.80\times$ baseline at +15 mm and $0.64\times$ at +20 mm. All three spectra peak in the first resolvable bin (3.3 Hz), so this is frequency resolution, not a detected physical mode. More importantly, PIMPLE marches the +20 mm case through the point where SIMPLE failed. The old “divergence = unsteadiness” interpretation is rejected.

This coarse transient mesh also fails to preserve the fine-mesh monotonic ride-height load trend (+15 mm carries anomalously more mean load than the coarse baseline). Therefore this experiment validates solver operability only; it does not validate the ride-height physics quantitatively. A defensible porpoising study requires a longer, fine-mesh URANS/DES run coupled to heave motion rather than a fixed-body force spectrum.

10 Credibility Discussion

- Geometry is a third-party reconstruction from imagery (CGTrader pre-season version); deviation from the real car is unknown;
- No wind-tunnel or track correlation data; all conclusions are trends;

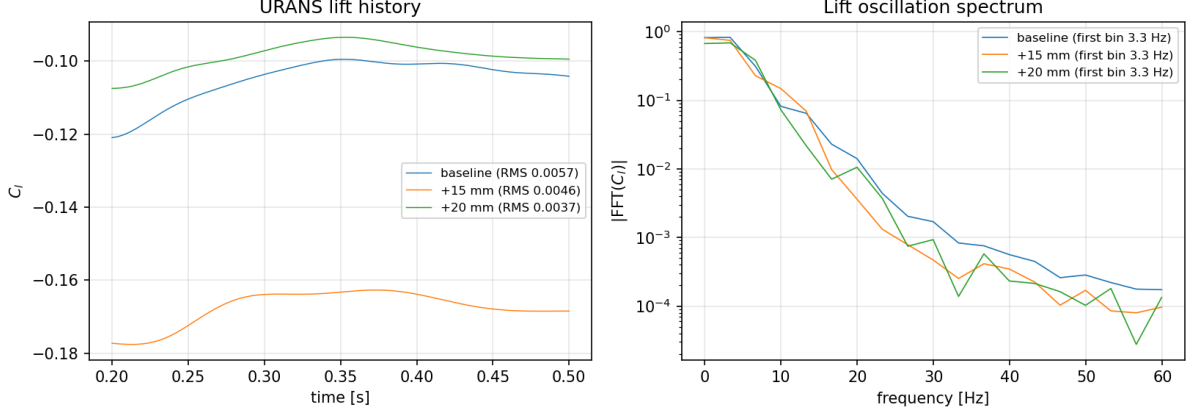


Figure 16: URANS lift history and spectrum: baseline, +15 and +20 mm (coarse mesh).

- First-order schemes (numerical diffusion quantified: C_l inflated $\approx 6\%$, see the second-order section), high y^+ , and closed cooling are the three systematic simplifications;
- Failure registry: mesh-fine (OOM); early kepsilon failures were all configuration errors (dimension names/dictionary keys/file headers), not physical conclusions; the first rough05/rough10 attempts failed numerically — initially misdiagnosed as roughness-constraint violations, later overturned by a compliance audit (violating faces $\leq 0.02\%$) and successful reruns under the stable recipe.

11 Conclusions and Future Work

At trial scale (4.35M cells, first order, high y^+), this campaign reproduced the methodology of the three reference works, completed an initial eight- dimensional sensitivity campaign, and added six follow-up analyses (wake, lap-time mapping, telemetry, corrected full-car yaw, attitude/contact geometry and URANS), for 37 usable simulation cases in total. Four additional v1 yaw cases are retained only in the failure registry and excluded from conclusions:

- **Methodology:** mesh-family study (honest conclusion: asymptotic range not reached), determinism verification, block-bootstrap uncertainty ($CI_{95} = 3\text{--}5\%$), first-order numerical-diffusion quantification (C_l inflated $\approx 6\%$);
- **Physical sensitivities:** roughness gains propagate through the front-tyre wake (47% carried by the front tyres); turbulence-model local differences are masked by whole-car coefficients (RKE local ΔC_p reaches $4\times$ the strongest roughness effect);
- **2026 active-aero approximation:** six-point rear-wing polar (no stall through -12° , unload side capped by floor upwash), front wing as a one-way balance lever (4° swings the front axle by 16 pp; the reverse direction stalls in ground effect), two-wing superposition holds, Straight-Mode balance cost -5.4 pp;
- **Geometry detail:** tyre contact plinths give the largest single-item L/D gain ($+11\%$);
- **Ground effect:** ride-height cliff (floor load -64% within 5 mm) and the steady-solver operability edge (divergence at +20 mm).

The production phase (32–64 GB, 25M+ cells, second order, wall-resolved wings) can reuse this pipeline wholesale; priority re-checks: the +15 mm-and-above ride region (URANS/DES), mesh convergence of the front-wing ground-effect stall, and Straight-Mode switch transients.

A All-Variant Results Table

Variant	Category	C_d	C_l	front%
Baseline		0.2413	-0.2498	24.9
Repeat run	Determinism	0.2413	-0.2498	24.9
Mesh 0.59M	Mesh family	0.3190	-0.1342	33.1
Mesh 2.0M	Mesh family	0.2598	-0.2522	36.6
Second-order momentum	Numerics	0.2369	-0.2357	24.5
Roughness 0.5mm	Wall	0.2437	-0.2697	23.1
Roughness 1.0mm	Wall	0.2457	-0.2851	23.7
Low- y^+ layers	Wall	0.2339	-0.2490	28.2
SST $a_1=0.31$	Turbulence	0.2417	-0.2586	28.7
Realizable $k-\epsilon$	Turbulence	0.2305	-0.2569	23.9
RW +8°	Active aero	0.2328	-0.2266	34.5
RW +4°	Active aero	0.2340	-0.2310	33.1
RW -4°	Active aero	0.2419	-0.2506	25.2
RW -8°	Active aero	0.2505	-0.2642	21.2
RW -12°	Active aero	0.2591	-0.2785	16.6
FW +4°	Balance	0.2408	-0.2079	8.6
FW -4°	Balance	0.2437	-0.2328	22.2
Both wings Straight-Mode	Active aero	0.2321	-0.1857	19.5
Tyre plinth 5mm	Geometry detail	0.2306	-0.2468	25.7
Tyre plinth 10mm	Geometry detail	0.2300	-0.2540	24.1
Tyre plinth 15mm	Geometry detail	0.2265	-0.2609	25.1
Raised +10mm	Ride height	0.2309	-0.2222	27.5
Raised +15mm	Ride height	0.2079	-0.1636	20.6
Plinth + raised 10mm	Combination	0.2197	-0.2352	—
Balance-neutral Straight Mode	Design closure	0.2324	-0.1932	23.9
Yaw -3° v2 (half-equivalent)	Yaw	0.2768	-0.1985	37.2
Yaw 0° v2 (half-equivalent)	Yaw control	0.2769	-0.2089	25.7
Yaw +3° v2 (half-equivalent)	Yaw	0.2754	-0.1867	38.0
Rake -0.3° (nose-down)	Attitude	0.2318	-0.2212	25.6
Rake +0.3° (nose-up)	Attitude	0.2273	-0.2005	34.1
Tyre squish 2mm	Geometry detail	0.2263	-0.2503	25.2
Tyre squish 4mm	Geometry detail	0.2266	-0.2527	24.7
Tyre squish 6mm	Geometry detail	0.2240	-0.2449	25.3
Tyre squish 8mm	Geometry detail	0.2250	-0.2534	25.9
URANS baseline (coarse)	Transient	0.2993	-0.1048	30.1
URANS +15mm (coarse)	Transient	0.3080	-0.1674	28.6
URANS +20mm (coarse)	Transient	0.3090	-0.0985	8.4
Raised +20mm	Ride height	Steady solver diverged $\times 2$ (operability edge)		

B Case Configuration Summary

OpenFOAM 14; foamRun incompressibleFluid; $k-\omega$ SST; 800 iterations; 4 vCPU EPYC-Turin / 15 GB. Complete dictionaries and solver settings are archived with the repository.